JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON SATURDAY 24th JANUARY 2026) **TIME : 9:00 AM TO 12:00 Noon**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. If the function $f(x) = \frac{e^x\left(e^{\tan x - x} - 1\right) + \log_e(\sec x + \tan x) - x}{\tan x - x}$ is

Continuous at x = 0, then the value of f(0) is equal to

(1) 2 (2) $\frac{2}{3}$

(3) $\frac{1}{2}$ (4) $\frac{3}{2}$

Ans. (4)

2. Let a circle of radius 4 pass through the origin O, the points A $\left(-\sqrt{3}a, 0\right)$ and B $\left(0, -\sqrt{2}b\right)$, where a and b are real parameters and ab ≠ 0. Then the locus of the centroid of ΔOAB is a circle of radius

(1) $\frac{5}{3}$ (2) $\frac{7}{3}$

(3) $\frac{8}{3}$ (4) $\frac{11}{3}$

Ans. (3)

3. Let the lines L_1: $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda\left(2\hat{i} + 3\hat{j} + 4\hat{k}\right)$, $\lambda \in \mathbb{R}$ and L_2 : $\vec{r} = \left(4\hat{i} + \hat{j}\right) + \mu\left(5\hat{i} + 2\hat{j} + \hat{k}\right)$, $\mu \in \mathbb{R}$, intersect at the point R. Let P and Q be the points lying on lines L_1 and L_2, respectively, such that $\left|\overrightarrow{PR}\right| = \sqrt{29}$ and $\left|\overrightarrow{PQ}\right| = \sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(QR)^2$ is equal to

(1) 340 (2) 360

(3) 320 (4) 348

Ans. (2)

4. Let 729, 81, 9, 1, …. be a sequence and P_n denote the product of the first n terms of this sequence.

If $2\sum_{n=1}^{40}\left(P_n\right)^{\frac{1}{n}} = \frac{3^\alpha - 1}{3^\beta}$ and gcd (α, β) = 1, then α + β is equal to

(1) 73 (2) 74

(3) 75 (4) 76

Ans. (1)

5. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that $\left|\vec{d} - \vec{a}\right| = \sqrt{11}, \left|\vec{c} \times \vec{d}\right| = 3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

(1) 11

(2) 3

(3) 0

(4) 1

Ans. (3)

6. If the domain of the function

$f(x) = \log_{(10x^2 - 17x + 7)}(18x^2 - 11x + 1)$

is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then 90(a + b + c + d + e) equals:

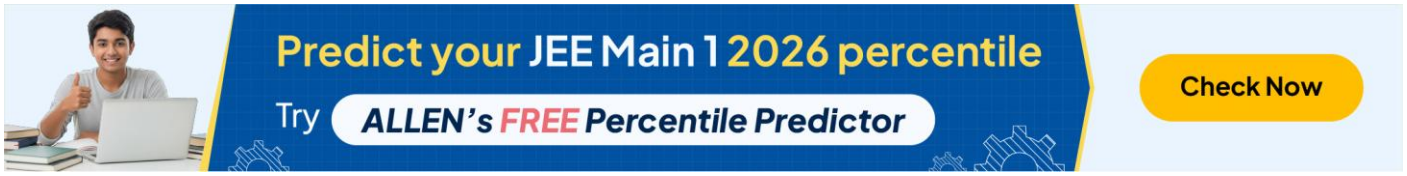
(1) 170

(2) 177

(3) 307

(4) 316

Ans. (4)



7. Let each of the two ellipses $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$, (A < B) have eccentricity $\frac{4}{5}$. Let the lengths of the latus recta of E_1 and E_2 be ℓ_1 and ℓ_2, respectively, such that $2\ell_1^2 = 9\ell_2$. If the distance between the foci of E_1 is 8, then the distance between the foci of E_2 is

(1) $\frac{96}{5}$ (2) $\frac{32}{5}$

(3) $\frac{16}{5}$ (4) $\frac{8}{5}$

Ans. (2)

8. The value of $\frac{\sqrt{3}\,\text{cosec}20° - \sec 20°}{\cos 20° \cos 40° \cos 60° \cos 80°}$ is equal to

(1) 32 (2) 16

(3) 64 (4) 12

Ans. (3)

9. Let $S = \left\{ z \in \mathbb{C} : \left|\frac{z-6i}{z-2i}\right| = 1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5} \right\}$.

Then $\sum_{z \in s} |z|^2$ is equal to

(1) 398 (2) 413

(3) 423 (4) 385

Ans. (4)

10. If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then

$\sin 7x \left(\cos\frac{13x}{2} + \sin\frac{13x}{2}\right) +$

$\cos 7x \left(\cos\frac{13x}{2} - \sin\frac{13x}{2}\right)$ is equal to

(1) $\frac{4}{\sqrt{26}}$ (2) $\frac{6}{\sqrt{26}}$

(3) $\frac{1}{\sqrt{13}}$ (4) $\frac{5}{\sqrt{13}}$

Ans. (3)

11. Let $f(t) = \int \left(\frac{1 - \sin(\log_e t)}{1 - \cos(\log_e t)} \right) dt, t > 1$.

If $f(e^{\pi/2}) = -e^{\pi/2}$ and $f(e^{\pi/4}) = \alpha e^{\pi/4}$, then α equals

(1) $-1-\sqrt{2}$

(2) $-1-2\sqrt{2}$

(3) $1+\sqrt{2}$

(4) $-1+\sqrt{2}$

Ans. (1)

12. Let R be a relation defined on the set $\{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$ by

$R = \{((a, b), (c, d)) : 2a + 3b = 3c + 4d\}$.

Then the number of elements in R is

(1) 6 (2) 18

(3) 12 (4) 15

Ans. (3)

13. Let A(1, 0), B(2, –1) and $C\left(\frac{7}{3}, \frac{4}{3}\right)$ be three points.

If the equation of the bisector of the angle ABC is $\alpha x + \beta y = 5$, then the value of $\alpha^2 + \beta^2$ is

(1) 8 (2) 5

(3) 13 (4) 10

Ans. (4)

14. Let $S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots$ up to 13 terms.

If $13S = \frac{2^k}{n!}$, $k \in N$, then

$n + k$ is equal to

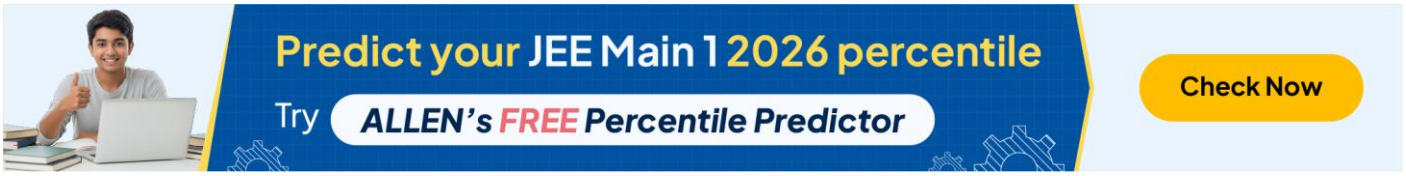
(1) 51

(2) 52

(3) 49

(4) 50

Ans. (3)



15. Consider an A.P.: $a_1, a_2, ..., a_n$; $a_1 > 0$. If $a_2 - a_1 = \frac{-3}{4}$, $a_n = \frac{1}{4}a_1$, and $\sum_{i=1}^{n} a_i = \frac{525}{2}$, then $\sum_{i=1}^{17} a_i$ is equal to

(1) 476 (2) 952

(3) 238 (4) 136

Ans. (3)

16. Let $\alpha, \beta \in \mathbb{R}$ be such that the function

$$f(x) = \begin{cases} 2\alpha(x^2-2)+2\beta x & , x < 1 \\ (\alpha+3)x+(\alpha-\beta) & , x \geq 1 \end{cases}$$

be differentiable at all $x \in \mathbb{R}$. Then $34(\alpha + \beta)$ is equal to

(1) 84 (2) 48

(3) 36 (4) 24

Ans. (2)

17. From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of getting at least 7 defective bulbs is :-

(1) $\frac{7}{10^7}$ (2) $\frac{81}{10^8}$

(3) $\frac{67}{10^8}$ (4) $\frac{73}{10^8}$

Ans. (4)

18. The mean and variance of a data of 10 observations are 10 and 2, respectively. If an observations α in this data is replaced by β, then the mean and variance become 10.1 and 1.99, respectively. Then $\alpha + \beta$ equals.

(1) 10 (2) 15

(3) 5 (4) 20

Ans. (4)

19. Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2$, $x + y = 8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y = x^2 + 2$, $y^2 = x$, $x = 2$, and y-axis that lies in the first quadrant. Then $A_1 - A_2$ is equal to

(1) $\frac{2}{3}\left(2\sqrt{2}+1\right)$ (2) $\frac{2}{3}\left(4\sqrt{2}+1\right)$

(3) $\frac{2}{3}\left(\sqrt{2}+1\right)$ (4) $\frac{2}{3}\left(3\sqrt{2}+1\right)$

Ans. (1)

20. The number of the real solutions of the equation :

$x|x+3|+|x-1|-2=0$ is

(1) 3 (2) 2

(3) 5 (4) 4

Ans. (1)

SECTION-B

21. Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right)dt = 4\alpha f(x)$.

If $y = f(x)$ is a standard parabola passing through the points (2, 1) and $(-4, \beta)$,

Then β^{α} is equal to _____.

Ans. (64)

22. Let a line L passing through the point P(1, 1, 1) be perpendicular to the lines $\frac{x-4}{4} = \frac{y-1}{1} = \frac{z-1}{1}$ and $\frac{x-17}{1} = \frac{y-71}{1} = \frac{z}{0}$. Let the line L intersect the yz-plane at the point Q. Another line parallel to L and passing through the point S(1,0, –1) intersects the yz-plane at the point R. Then the square of the area of the parallelogram PQRS is equal to _____.

Ans. (6)



23. The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is _____

Ans. (42)

24. Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{|t+1|}{t^2}$, $t < 0$, is strictly decreasing. Then the local maximum value of the function $g(x) = 2\log_e(x-2) + \alpha x^2 + 4x - \alpha$, $x > 2$, is ____

Ans. (4)

25. The number of 3×2 matrices A, which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of A^TA is 5, is _____

Ans. (312)



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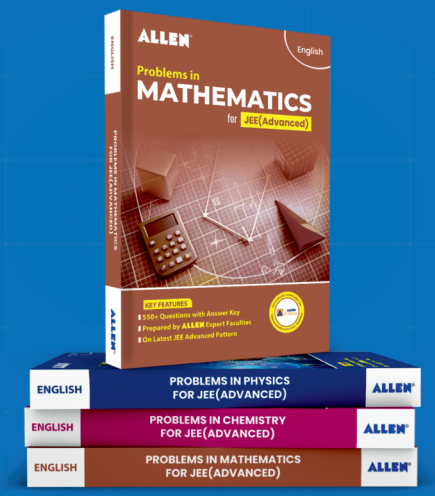


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